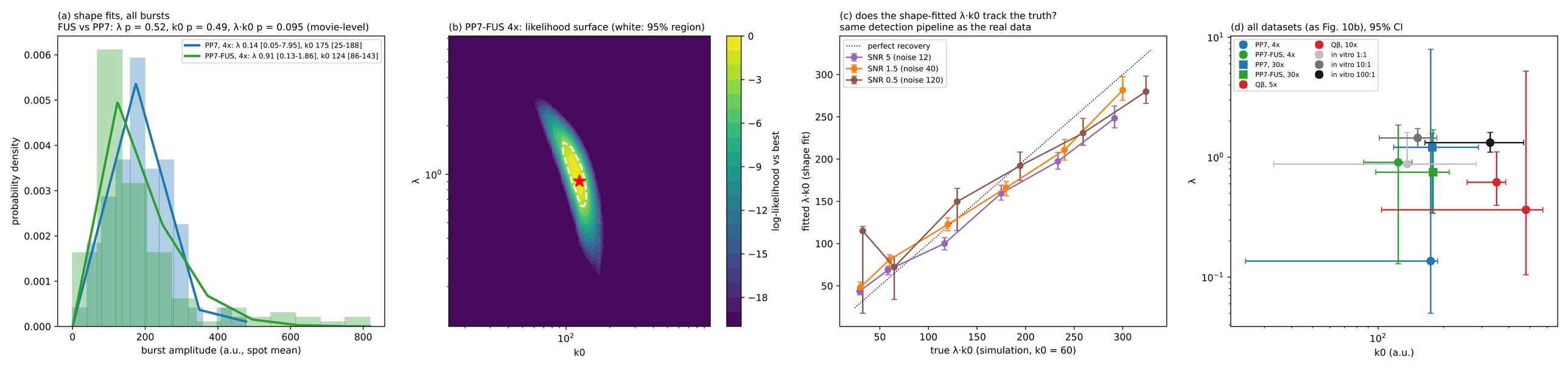
(a) shape fits, all bursts
FUS vs PP7: λ p = 0.52, k0 p = 0.49, λ·k0 p = 0.095 (movie-level)
PP7, 4x: λ 0.14 [0.05-7.95], k0 175 [25-188]
PP7-FUS, 4x: λ 0.91 [0.13-1.86], k0 124 [86-143]
probability density
burst amplitude (a.u., spot mean)
(b) PP7-FUS 4x: likelihood surface (white: 95% region)
λ
k0
log-likelihood vs best
(c) does the shape-fitted λ·k0 track the truth?
same detection pipeline as the real data
perfect recovery
SNR 5 (noise 12)
SNR 1.5 (noise 40)
SNR 0.5 (noise 120)
fitted λ·k0 (shape fit)
true λ·k0 (simulation, k0 = 60)
(d) all datasets (as Fig. 10b), 95% CI
PP7, 4x
PP7-FUS, 4x
PP7, 30x
PP7-FUS, 30x
Qβ, 5x
Qβ, 10x
in vitro 1:1
in vitro 10:1
in vitro 100:1
λ
k0 (a.u.)